

OpenGL Coursework 2 Prototype

This project is a 3D prototype built in C++ using OpenGL.

It features an environment that has two textures, one with grass and another with my signature, and finally, camera movement.

Video link -> <https://youtu.be/WzwAMtOtyyk>

GitHub link -> <https://github.com/sniffleslou-lab/comp3016CW2silver>

Dependencies

- GLFW (WINDOW + input.)
- GLEW (OpenGL function)
- STB Image (textures)
- GLM (maths Library)

Game Description

The user can freely move around a 3D scene:

- W/A/S/D for movement
- Mouse for the camera

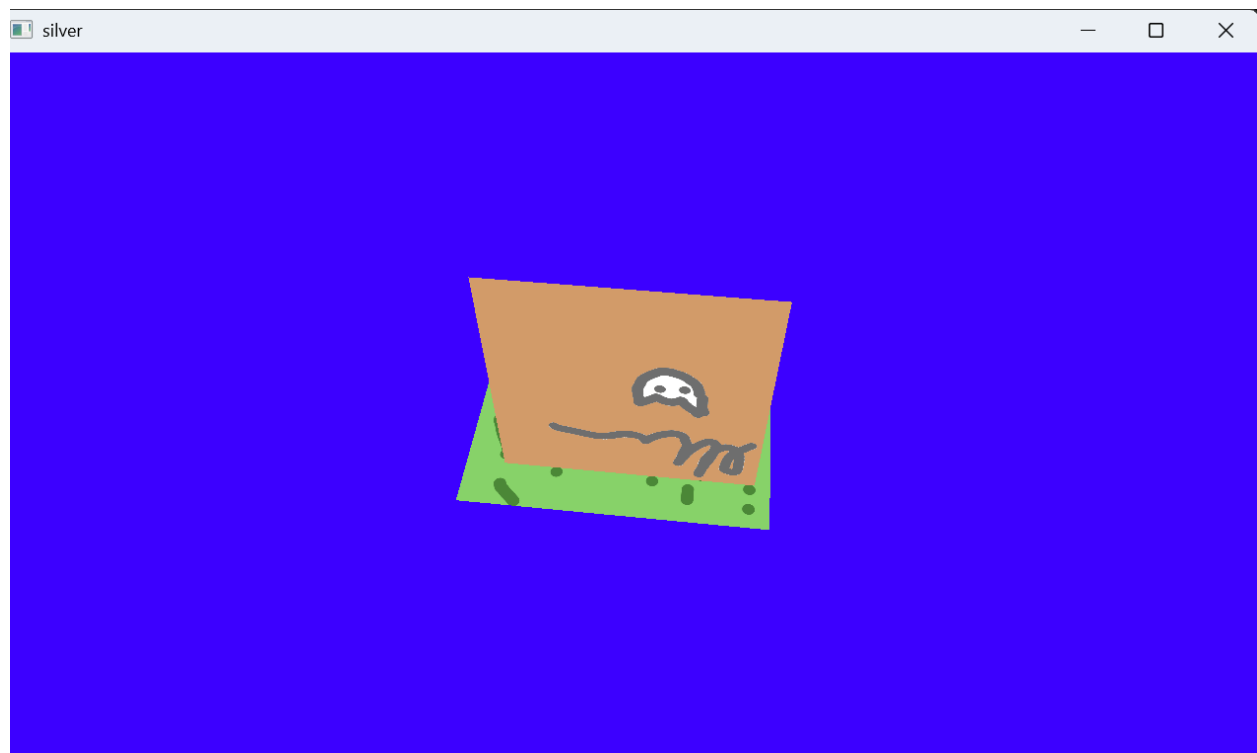
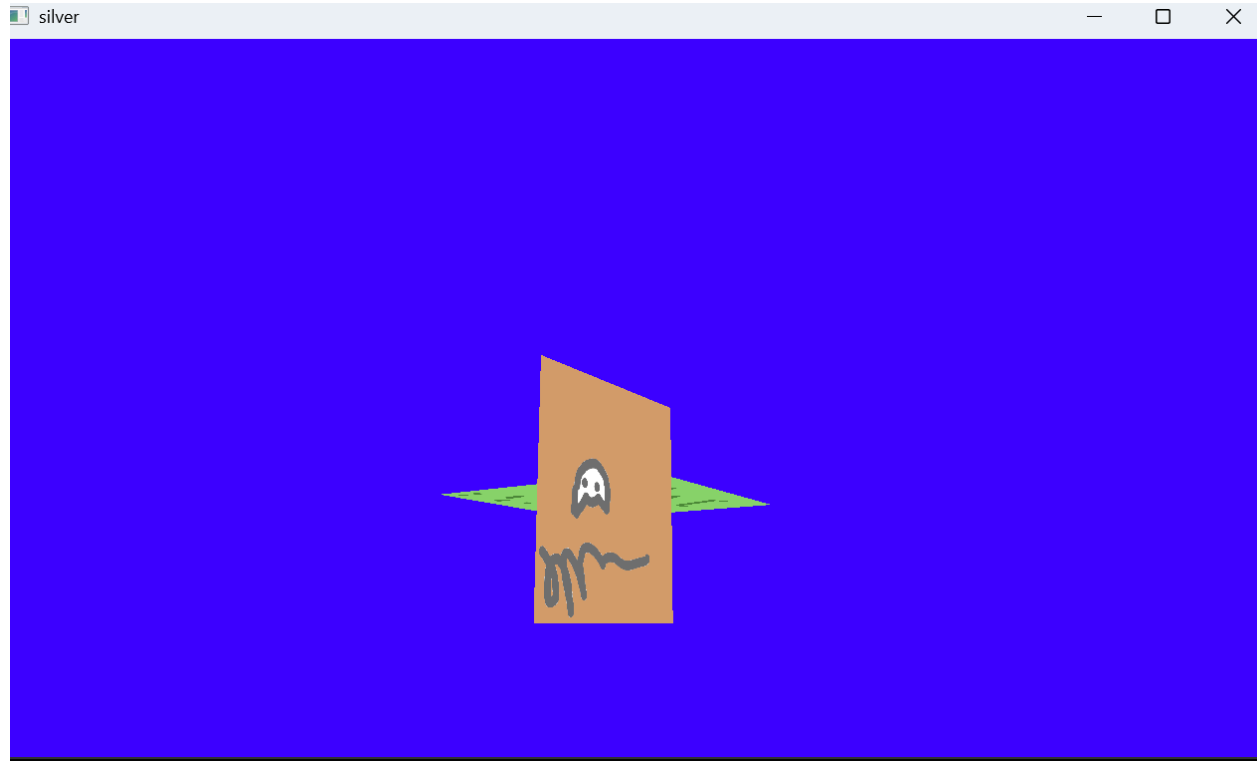
The scene contains

- Grass plane
- Signature quad
- 3d environment with perspective camera

Mechanics are simple. The user is able to move around with the camera. This is done as the movement and camera are tied.

Rendering is interrogated using VAOs and VBOs, with textures sampled using 'sampler2D' with the use of the fragment shader.

Screenshots of game running



Code

I began the project using the code I previously produced and I used the provided solution from lab 5 as a reference. These were the foundation to make sure the project has set up the provided libraries correctly and allow the OpenGL window to launch. I rewrote and tweaked the code to fit the requirements for the project.

Loadshaders.cpp was another piece of code I was stuck on during the time of the lab work, and I based my current Loadshaders on the solution that was given. I integrated it into my project.

The rest of the project. Gameplay logic, scene, textures and the camera were implemented by me.

Testing

Due to the nature of the end project, testing is done by checking that shaders are loading and moving the camera.

I always had issues with visual study with the directory of the the project, so I always used something on the lines like

```
' char cwd[256]; _getcwd(cwd, 256); std::cout << "Working directory: " << cwd << std::endl;'
```

Which will give the work directory of my code, I will often use code lines like these to give errors in the terminal to check if stuff is connected or working.

```
'std::cout << "texture failed to load : " << filepath << std::endl;'
```

Evaluation

The end project in light wording is 'rather bad'. I had ambitions to make something big, but the changing seasons hit me personally hard, and I decided I would rather just pass with the minimum requirements, being safe with what I have, instead of trying to make something big and realising I am unable to submit anything due to an unfinished mess.

If given the chance to redo this project, let's say in the summer, if I fail and I mess up, I would probably aim to be ambitious and integrate more features into the project, making the game have grass, trees and a feature.